



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to ASR HI-TECH FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

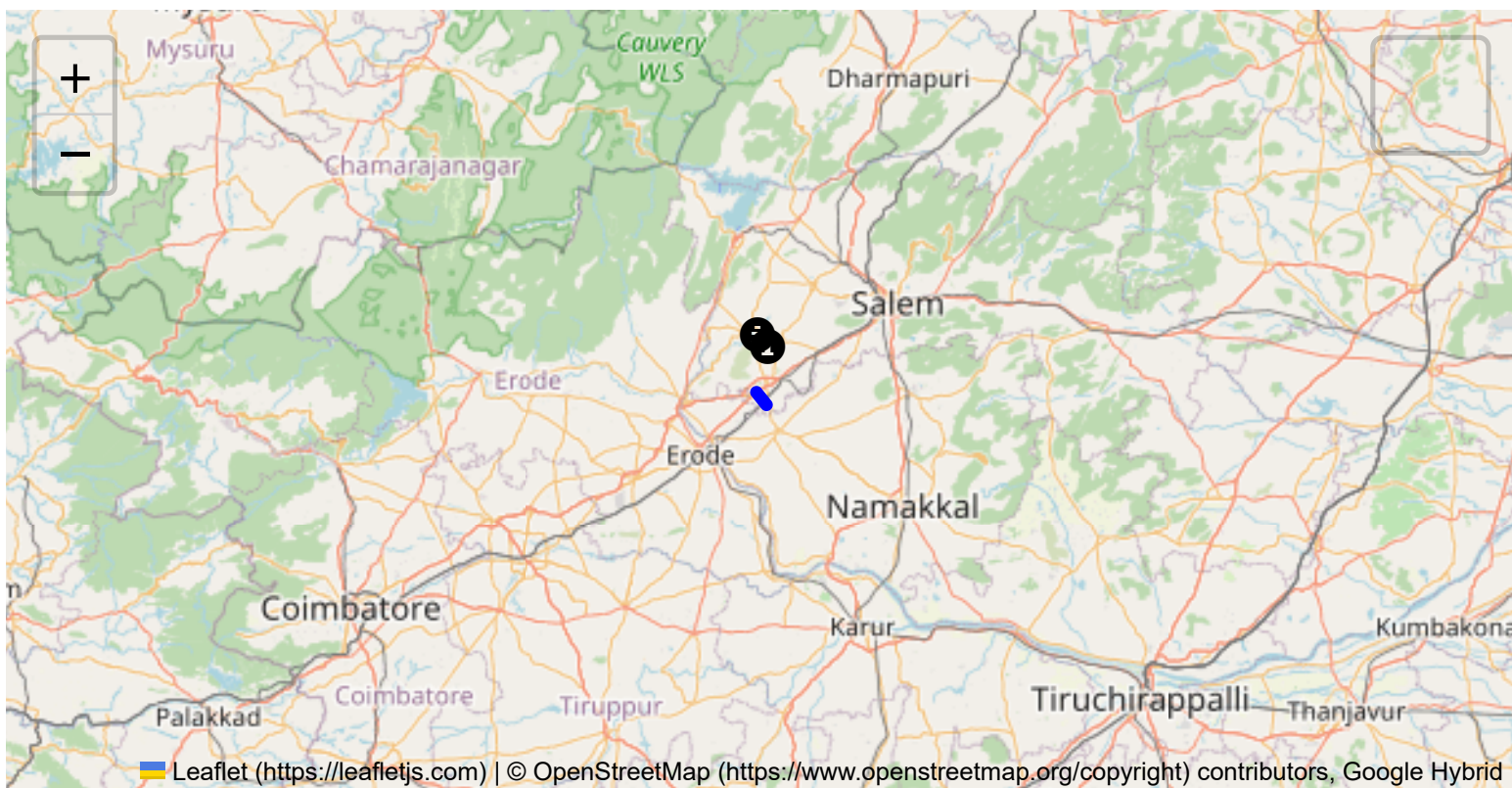
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

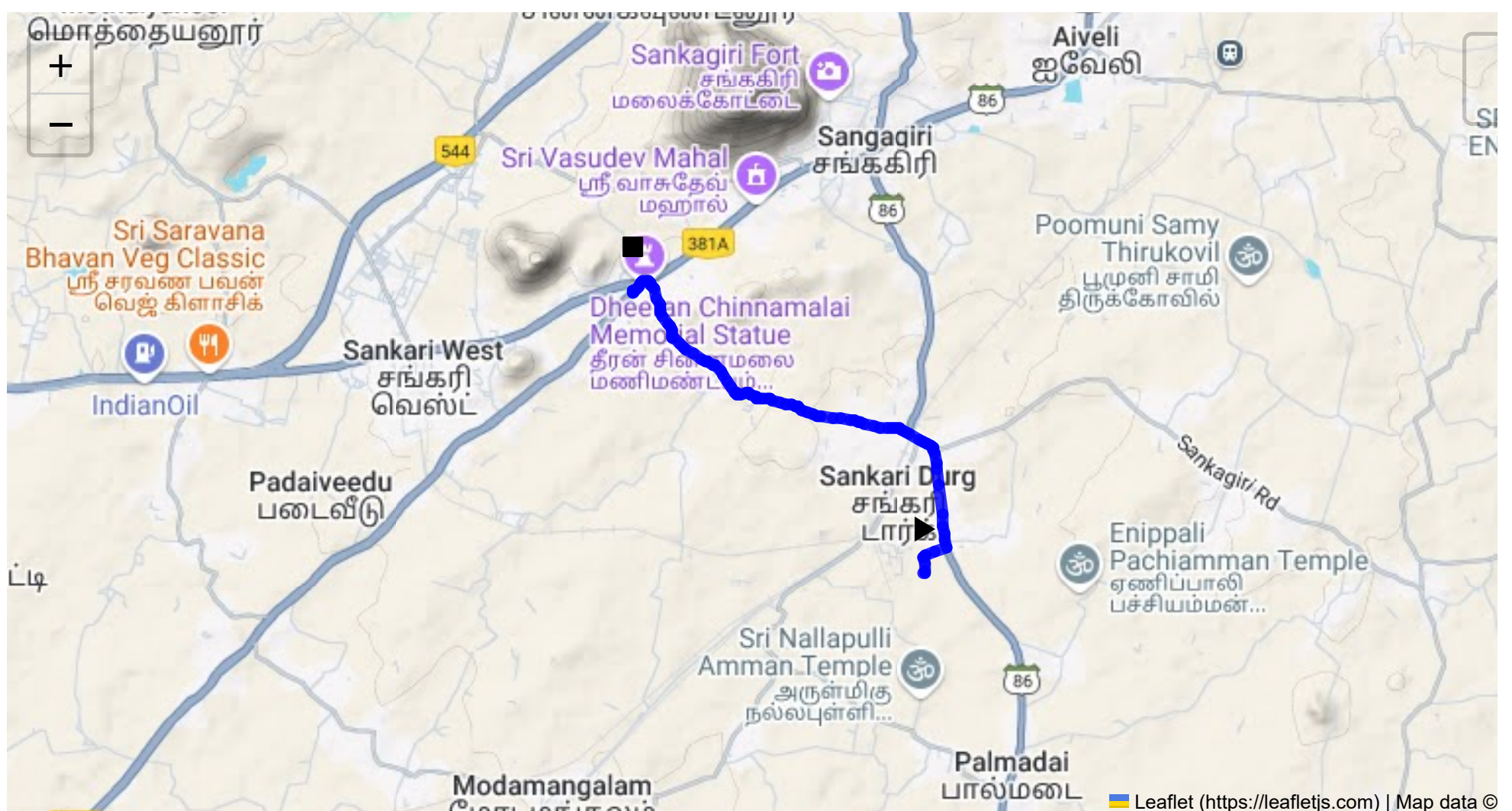
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 5.00 km
Estimated Duration: 0.2 hours
Adjusted Duration (Heavy Vehicle): 0.2 hours
Start: (11.4381, 77.8734)
End: (11.462413, 77.847681)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri, Tamil Nadu to FR6X+R6C, Muniyappanpalayam, Tamil Nadu spans approximately 5 kilometers, typically taking around 10 minutes for heavy vehicles carrying hazardous materials. It is a relatively short route that likely involves local roads and possibly a mix of rural and semi-urban areas, common in regional routes within Tamil Nadu.

Weather Conditions

- **Typical Weather:** The region experiences a tropical climate with high temperatures, especially between March and June. Monsoon rains occur from October to December.
- **Weather-related Hazards:** During the monsoon, heavy rains can lead to waterlogging and reduced visibility. High temperatures may also cause road surface softening or tire issues.

Traffic Patterns

- **Peak Hours:** Local traffic generally peaks in the morning (8-10 AM) and evening (5-7 PM) due to commuting.
- **Congestion:** The route might not experience significant congestion due to its short length and probable rural nature, but traffic can increase during market days or local events.

Road Quality and Infrastructure

- **Assessment:** The roads are likely to be narrow, with potential potholes and limited lighting. Road quality may vary, with some stretches possibly lacking proper signage and barriers.
- **Infrastructure:** Limited amenities like service stations or rest areas could be present.

Alternative Routes for Emergencies

- **Suggestions:** While local roads may offer limited alternatives due to their layout, drivers can use connecting roads to nearby towns like Namakkal or Salem as potential detour options in emergencies.

Local Regulations on Hazardous Material Transport

- **Regulations:** Transporting hazardous materials requires compliance with national regulations, including appropriate signage, container integrity checks, and driver certifications.

Historical Incidents

- **Overview:** There is limited publicly available data on specific incidents along this short route. However, regional data from Tamil Nadu often reports issues with truck accidents and chemical spills in more industrial areas.

Environmental Considerations

- **Sensitive Areas:** The region may have agricultural zones. Care should be taken to prevent spills that could contaminate soil or water, affecting local farming activities.

Communication Coverage

- **Dead Zones:** Mobile network coverage in rural Tamil Nadu can be spotty. Drivers should ensure they have backup communication methods, especially in sparsely populated areas.

Emergency Response Times

- Estimates:** Emergency response times can vary but generally expect 20-30 minutes in rural areas due to the local availability of services.

Overall Summary of Risk Assessment

- Risk Level:** Moderate risk due to potential weather hazards, road quality concerns, and infrastructure limitations.
- Recommendations:** Regular vehicle maintenance checks, adhering to safety regulations, and having alternate communication methods are essential. Drivers should also be familiar with emergency procedures and potential detour routes.

Given the local conditions and typical rural infrastructure, it is crucial for truck drivers to be vigilant and prepared for unexpected developments, especially when transporting hazardous materials. Regular communication with dispatch and local authorities is key to ensuring safety along the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4622233, 77.8479478	10 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.46307, 77.84941	30 KM/Hr
6	Turn	Medium	11.46318, 77.84913	30 KM/Hr
7	Turn	Medium	11.46312, 77.84886	30 KM/Hr
8	Turn	High	11.46222, 77.84795	15 KM/Hr
9	Turn	High	11.46226, 77.84791	15 KM/Hr
10	Turn	High	11.46230, 77.84795	15 KM/Hr
11	Turn	Medium	11.46241, 77.84786	30 KM/Hr

Risk Assessment - Elevation Changes

	Risk Type	Specific Risk	Risk Level	Coordinates	Speed Limit
0	Elevation Change	Descent	Medium	11.46081632613708, 77.85028396163287	30 KMS/Hr
1	Elevation Change	Descent	Medium	11.46317981267479, 77.8491307493013	30 KMS/Hr

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4622233, 77.8479478



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46222, 77.84795



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46226, 77.84791



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46230, 77.84795



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46241, 77.84786
